

Yiunfan Hu

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EDUCATION

BSc. Mechanical Engineering at Stony Brook University

University Scholar | Dean's list (all semesters) | GPA: 3.98 | U4 Standing

Stony Brook, NY

Jun 2024–May 2027

(expected)

Upper Division MEC Courses: [364] *Introduction to Fluid Mechanics*, [363] *Mechanics of Solids*, [301] *Thermodynamics*, [325] *Manufacturing Processes*, [316] *Instrumentation and Solid Mechanics Lab*, [310] *Introduction to Machine Design*

* **In progress Spring 2026:** [410] *Design of Machine Elements*, [305] *Heat and Mass Transfer*, [320] *Numerical Methods in Engineering Design and Analysis*, [317] *Thermal Sciences and Fluid Mechanics Lab*

SKILLS

CAD	Autodesk Inventor, Autodesk Fusion, Onshape, SolidWorks
Mechanical	DFM, Rapid Prototyping, 3D Printing, CNC Machining, Manual Machining, Laser Cutting, Plasma Cutting
Electrical	Arduino, Soldering
Software	C, C#, HTML/CSS/JS, MATLAB, Python

EXPERIENCE

Reactive Collision UAV | Research Member at Prof. William Stewart's Soft Flyers Group

Stony Brook, NY

Jun 2025–present

- Assisted in designing a collision-resistant fixed-wing quadrotor with spring-loaded pivoting wings.
- Developed a launcher capable of accelerating the 1 kg UAV to 20 m/s.
- Fabricated major components such as the wings using a CNC router.

Stony Brook Robotics Team – MATE ROV | Float Team Lead

Stony Brook, NY

Sept 2025–present

- Designed and built a profiling float for the 2026 Competition in Newfoundland, Canada
- Developed a hydraulic actuator system and contributed to arm design on the main ROV

Shape Memory Alloy (SMA) Springs | Research Member in a joint project with Interacting Robotic Systems Laboratory, Soft Flyers Group, and Yao Research Group

Stony Brook, NY

May 2024–present

- Developed 2-DOF joint with Arduino control system. Poster presentation in Summer Research Symposium at Stony Brook, August 2024.
- Designed and fabricated a test platform to control SMA-spring heating and quantitatively measure resulting displacement and temperature change.
- Experimentally validated theoretical models from literature and extended simple single-spring models to novel multi-spring networks.

PROFESSIONAL TRAINING

NASA L'SPACE Proposal Writing and Evaluation Experience Academy | Participant – Team Role: Engineer

Virtual

May–Aug 2025

- Workforce preparation academy dedicated in writing and evaluating professional technical proposals taught by NASA scientists, engineers, and managers.
- Developed a proposal focused on the laser ignition system for a rotating detonation rocket engine.
- Served on a review panel for NASA solicitation responses and gained hands-on experience in evaluating proposals based on solicitation rubric and identifying strengths and weaknesses.

EXTRACURRICULAR PROJECTS

Combat Robotics at National Havoc Robot League (NHRL) | Team Captain

Norwalk, CT

Jun 2023–present

(Jun 2023–Jun 2024)

- Leading a team building combat robots in the 12lb-weight-class at NHRL.

12lb Vertical Spinner Robot

- Designed and built a combat robot from scratch over 4 months with a hard competition deadline.
- Obtained rich experience in system integration, control systems, and metal manufacturing.
- Competed in September 2023 and June 2024.

12lb Differential Swerve Drive Robot

(Jun 2024–present)

- Developed a novel compact drive unit enabling omni-directional movement for small combat robots.
- Currently designing a robot to integrate the drive units into, expected to compete in 2027.

"Javelin" Compact Electric Scooter | Solo Maker

Stony Brook, NY

Oct 2024–present

- Designed and built one of the most compact electric scooters over 6 months under a \$1,000 budget.
- Optimized packaging and maximized space utilization in design.
- Built a custom 10s2p li-ion battery pack with a 10Ah capacity as an upgrade to the commercial battery pack.
- Currently in use for daily commute.